📘 1. ATOMIC STRUCTURE – DETAILED SHORT NOTES

1. Dual Nature of Matter

Light shows both wave and particle nature

de Broglie relation: lambda = h / mv

2. Photoelectric Effect

Light → electrons emit from metal surface

Important points:

Threshold frequency hoti hai

Intensity ↑ → number of electrons ↑

Frequency ↑ → energy ↑

3. Bohr Model

Electron fixed circular orbits me revolve karta hai

Energy levels: n = 1, 2, 3...

Energy of electron:

E = -13.6 / n^2 (in eV)

Radius:

r ∝ n^2

4. Quantum Numbers

Edit with WPS Office

n (principal) → shell

l (azimuthal) → subshell (s, p, d, f)

m (magnetic) → orientation

s (spin) → +1/2 or -1/2

5. Electronic Configuration

Aufbau principle → lower energy first

Pauli principle → max 2 electrons/orbital

Hund rule → maximum unpaired electrons

Exception examples: Cr, Cu

6. Heisenberg Uncertainty Principle

Position and momentum exact ek sath nahi pata

7. Orbitals

s → spherical

p → dumbbell

Edit with WPS Office